

JPMorgan Chase is the largest bank in the United States and one of the largest in the world. It processes hundreds of millions of transactions daily across retail banking, credit cards, investment accounts, and wire transfers. That amount of transactions makes reviewing every hundred million documents for potential fraud practically impossible. As such, starting in 2016 and accelerating in the 2020s, JPMorgan deployed a suite of AI systems, notably its Contract Intelligence (COiN) and broader fraud detection ML infrastructure.

JPMorgan employed numerous AI technologies across its systems, including machine learning and supervised and unsupervised models. Combining gradient boosting, random forests, and neural networks trained on historical labeled fraud cases. Where the supervised models learned patterns from fraud, such as flagged transaction amounts, merchant categories, geolocation anomalies, time-of-day patterns, and device fingerprints, while the unsupervised models identified unknown fraud typologies by detecting statistically anomalous behavior without needing a labeled example first.

JPMorgan also employed Graph Neural Networks, which map the relationships between entities such as accounts, IP addresses, devices, merchants, and individuals; as such, it can surface hidden connections between fraudsters that often operate in rings with shared phone numbers, addresses, or devices, connections that tabular ML models may miss.

Natural Language Processing models, with their analysis of unstructured data for linguistic signals associated with social engineering; Real-Time Scoring engines, which can score transactions in under 100 milliseconds; and behavioral biometric AI, which analyzes how a user types, swipes, and navigates the app, are some of the other forms of technology that were used by JPMorgan Chase.

With all these efforts, the benefits of AI systems have become apparent. From fraud loss reduction, where industry estimates suggest that major banks using ML-based detection reduce fraud losses by 20%-40%; false positive reduction, where AI reduces legitimate transactions being declined which is a critical customer experience metric; speed in reviewing the transactions, where AI can review in milliseconds transactions that would take hours for a person review, leaving them with the more ambiguous cases; the scalability of model the infrastructure, where it can be expanded from reviewing thousands to reviewing hundred of millions without a proportional headcount increase; adaptive learning, where models can be retained as fraud patterns evolve;

and synthetic identity fraud detection; where specifically GNN-based approaches can address synthetic identity fraud, in which fraudsters construct fictitious persons from real data fragments.

That said, these AI systems are not without their challenges. Some of the challenges that come with these models include: model explainability, as deep learning algorithms are notoriously opaque, which is a problem for regulators who require that adverse actions be explainable to customers, and is what led JPMorgan and its peers to invest in SHAP values and other explainability tools to generate rationale for why a transaction was flagged or an account was frozen; adversarial machine learning, where sophisticated fraud rings probe AI systems deliberately and identify their vulnerabilities to take the advantage, such as making small incremental changes to transactions, so as to stay below detection values; data quality and label noise, as fraud labels in training data are imperfect, some fraud goes unreported or undetected, degrading model precision, and creating systematic blind spots; regulatory and fair lending risk, as when fraud models inadvertently use features correlated with protected class characteristics, like zip codes as a proxy for race, the institution will face a disparate impact liability, so ongoing bias audits are required for ensuring model fairness and accuracy; operational integration complexities, as deploying real-time scoring requires deep integration with legacy core banking infrastructure, systems that were often built in COBOL (or Common Business-Oriented Language, which is a legacy English-like programming language designed in 1959 for business data processing, and has been considered outdated regardless of its high-demand) and were decades old, with the engineering lift required to retrofit AI into existing transaction pipelines being substantial costly; and model risk management, where the Federal Reserve's SR 11-7 guidance requires rigorous validation of all models used in risk decisions, meaning that independent model validation teams must assess performance, stability, and appropriateness, which could create governance overhead and slow deployment cycles.

While these challenges, and general challenges of transparency, fairness, and explainability that JPMorgan is facing, are ones that are too important to overlook, there's no denying that JPMorgan Chase's AI fraud infrastructure represents, arguably, the most mature development of this technology in financial services. With its combination of real-time ML scoring, GNN relationship mapping, and NLP analysis, the AI fraud infrastructure creates a defense-in-depth through overlapping detection layers, and the measurable benefits in loss prevention and operational efficiency are substantial.